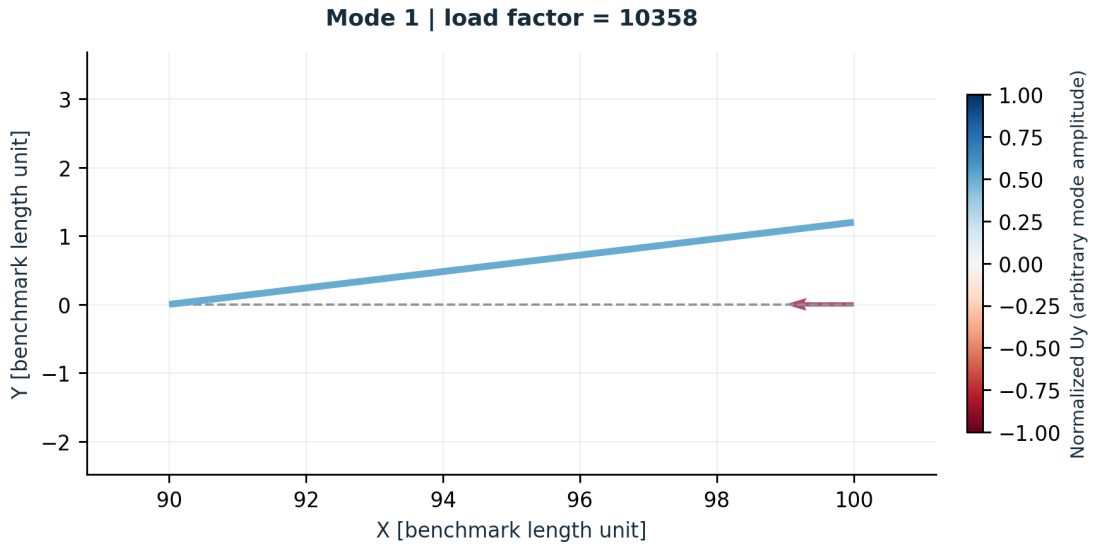
Single-element cantilever column buckling

MYSTRAN_sol105_buckling | SOL 105 | Independent analytical verification

One CBAR of length 10; E=1e7, A=0.5, I1=1/24, I2=104.16666666667.

Geometry / mesh	2 nodes; 1 elements. One CBAR of length 10; E=1e7, A=0.5, I1=1/24, I2=104.16666666667.
Material / section	E=1e+07, nu=0.33; density=0.1. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	All 6 DOFs at root GRID 1 fixed using permanent GRID constraints; unit axial compressive end force.
Applied load	Unit axial compressive tip force; Euler comparison uses the weak-axis second moment of area.



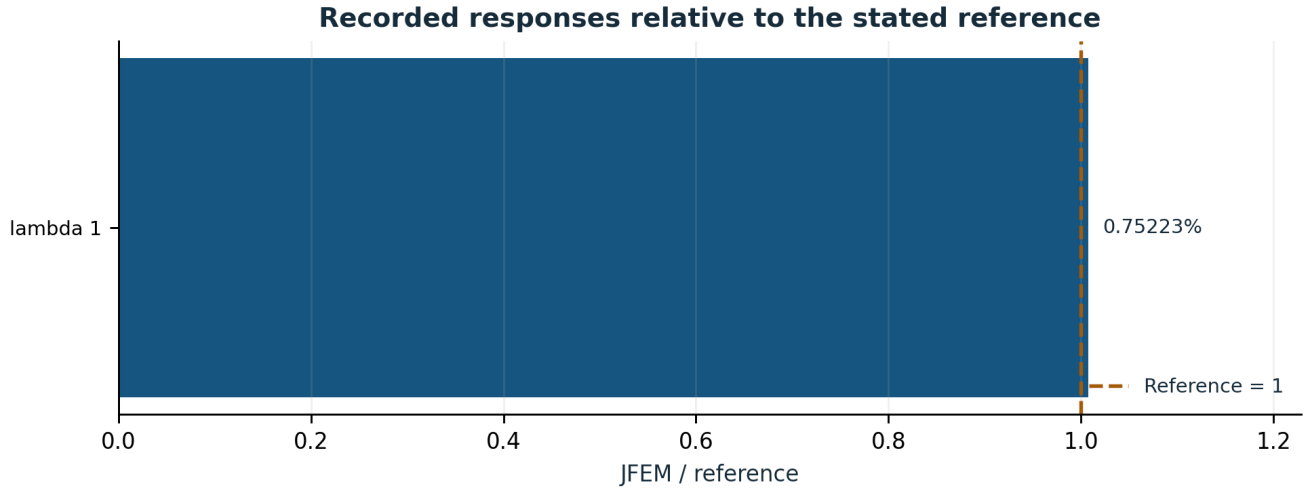
Shape amplified x1.2; grey = original. Eigenvector sign and amplitude are arbitrary.

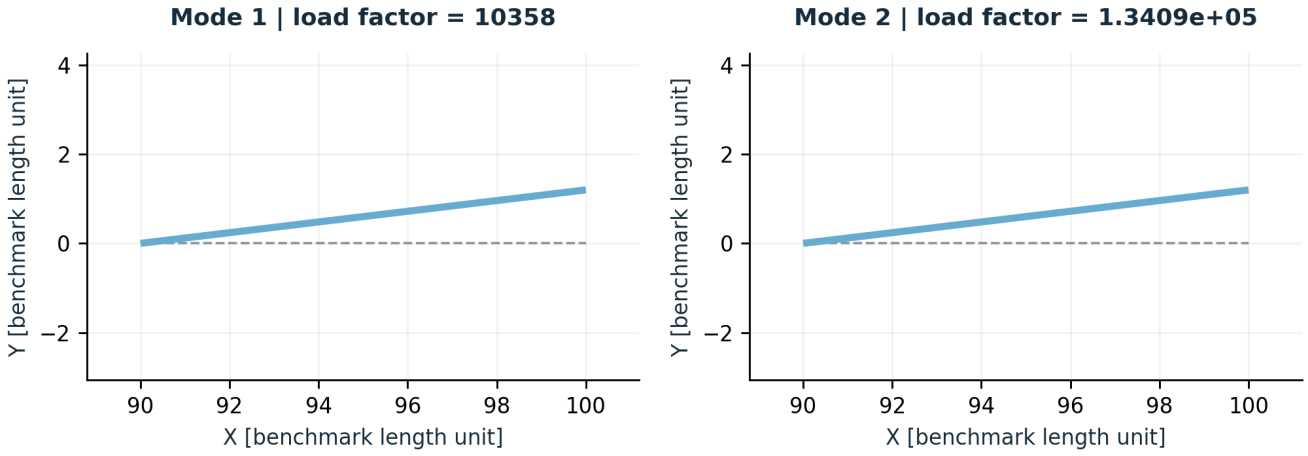
Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
lambda 1 [-]	10358	10281	0.75223%	PASS

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

Actual shipped mesh has one beam element. Euler value is the independent continuum target; finite-mesh error is retained.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MYSTRAN open-source test input provenance (MIT license); stored historical output values are regression targets only.
https://github.com/MYSTRANsolver/MYSTRAN

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv